

**Name of Staff:** Fredrick Onyango Awiti  
**Profession:** Application Geospatial Engineer  
**Name of Firm:** Ecospace Services Ltd  
**Date of Birth:** 07-Feb-1989  
**Nationality:** Kenyan

**Education:** **Oct. 2018 – Sep. 2020,** MSc Geographic Information System, the University of Nairobi.

**Oct. 2008 - Nov. 2013,** BSc Geomatic Engineering Geospatial Information System (Jomo Kenya University of Agriculture and Technology, Kenya)

**Professional Association:** Full Member of the Institution of Surveyors of Kenya.  
Project Management Institute (K) Member.  
Associate Member of the International GNSS Service (IGS.)

**Other training:** Continuous Operating Reference System (CORS) – Trimble Africa (Ghana).  
Remote Pilot License (RPL) – Drone Space (K).  
Drone Data Analysis using Trimble eCognition.  
Drone Data Processing and Analysis using DJI Terra and Pix4D Mapper – DJI Online.

**Countries of Work Experience:** Kenya, South Sudan, Somalia, Rwanda, Uganda, Tanzania

**Languages:**

| Language  | Reading   | Writing   | Speaking  |
|-----------|-----------|-----------|-----------|
| English   | Excellent | Excellent | Excellent |
| Kiswahili | Excellent | Excellent | Excellent |

**Employment Record:**

**Jan, 2019-Present** Managing Director & Geospatial Application Engineer, Ecospace Services Ltd, Nairobi.  
As Managing Director and Lead Geospatial Application Engineer at Ecospace Services Ltd, I provide both strategic direction and hands-on technical oversight across the company's surveying, sales, GIS, and geospatial application development operations. I am responsible for steering Ecospace's vision of delivering innovative, data-driven solutions for land management, environmental monitoring, and infrastructure development.

I oversee the full lifecycle of geospatial projects from conceptualization, planning, and data acquisition to analysis, visualization, and delivery. My duties include designing and

implementing data capture templates, geodatabases, and field workflows using tools such as QGIS, ArcGIS, PostgreSQL/PostGIS, and GeoServer. I also prepare comprehensive technical and financial tender documents, coordinate proposal submissions, and ensure compliance with regulatory and client requirements.

I manage and mentor multidisciplinary teams, ensuring efficient production, field data collection, and quality control. I also supervise the maintenance and calibration of equipment including GNSS Receivers, total stations, and drone platforms ensuring operational readiness, firmware updates, and equipment reliability.

Beyond technical delivery, I lead client engagement, product demonstrations, and professional training sessions to build capacity in modern surveying, remote sensing, and geospatial data management. I frequently conduct demonstrations on new technologies such as UAV mapping, photogrammetry, LiDAR integration, and AI-driven analytics, guiding clients to adopt tools best suited to their operational and strategic goals.

Strategically, I oversee business growth initiatives, partnership development, and digital transformation projects. Under my leadership, Ecospace has successfully executed projects across Kenya and the East African region, including GIS-based decision support systems, drone mapping for agriculture and infrastructure, land information management systems (LIMS), and environmental monitoring dashboards.

Through this dual role, I bridge executive management with deep technical proficiency ensuring that Ecospace remains at the forefront of geospatial innovation, delivering practical, scalable, and impactful solutions to clients in both public and private sectors.

**Aug 2018– Dec. 2018**

GIS & Quality Assurance Consultant, Locomizer Ltd, Nairobi, KENYA,

Manage, oversee, and supervise the creation of data capture templates, defining parameters, and conducting internal training support on what and how to capture the data. Conduct data validation and cross-checking to ensure proper and reliable data extraction from SQL databases; Contribute to the design, development, and maintenance of spatial data infrastructure and geodatabases to ensure smooth data flow to geoportals and other GIS solutions. Prepare and present weekly, monthly, quarterly, and yearly technical reports.

**Jul 2016 – Jul 2018**

Survey Projects Specialist, Oakar Services, Nairobi, KENYA,

As a project specialist, he worked closely with data processors, Client Deliver Consultants, and clients throughout the project lifecycle (preplanning, production, quality checking, fielding, and deliverable stages). He ensures that all objectives are understood by the team and are responsible for taking the lead in driving the project support team toward

the common goal of efficient production, seamless support, process adherence, and flawless delivery of data files within the agreed amount of time. He closely worked with the sales account executive to carry out demos and showcase the evolving new trends in surveying instruments and advise clients on the best products for their needs, ensured that the field equipment software is up to date and faulty ones were repaired, prepare technical presentations and demonstrations, prepare technical proposals and expressions of interest; Conduct training in GIS, GNSS & Remote Sensing; and Offer Technical Support to clients on GIS and Remote Sensing Software;

**May 2015 – June 2016**

Survey Solution Executive, Oakar Services, Nairobi, KENYA,

Closely worked with the sales account executive to do demos, showcase the evolving new trends in surveying instruments, and advise clients on the best products for their needs. He ensured that the field equipment software is up to date and faulty ones were repaired. Carried out internal training to the firm staff and externally to different organizations including; the Ministry of Land Department of Survey; SMEC International Kenya, Survey of Kenya (SoK), and Rural Electrification Authority (REA) among others about Geodetic GPS, Robotic Total Station, Handheld GPS and Survey Software like Carlson Survey CAD, Spectra Precision Survey Office (SPSO). He has been involved in conducting engineering, cadastral, topographical, and control survey for different organizations like; The Technical University of Kenya (TUK), The Nairobi University (UoN), Sogea Satom, ADCC International, Zhonghao Overseas, Kajiado County and Shymar Power among other.

**Oct. 2014 – Apr. 2015**

GIS Technician, Oakar Services, Nairobi, KENYA.

Carry out data collection using Handheld GPS and Smartphones; Digitization of satellite and scanned images using ArcGIS software; undertake data development activities by updating and maintaining files, personal geodatabases, and SQL databases; Produce and maintain high-quality cartographic maps using ArcGIS software; Conduct spatial data and satellite image analysis using ArcGIS and ERDAS Imagine; Provide client support on GIS, Remote Sensing software and GPS needs; Conduct training on ArcGIS, GPS, and Remote Sensing software; Develop customized training materials to target clients on ArcGIS and GPS; Undertake project management and implementation; Actively involved in business development through technical marketing, presentations, and demonstrations. Represent the company in technical workshops and seminars; and provide technical support to clients on GIS technologies.

**AMay 2013 – July 2013**

Graduate Trainee, Oakar Services Ltd, Nairobi, KENYA.

He was involved in GIS projects; digitizing, editing, publishing geospatial datasets, and mapping production. Participated in field data collection in preparation to build a GIS Database for the National Irrigation Board, providing support to clients on matters of GIS, GNSS equipment, Optical solutions, and processing software.

- Jan. 2012 – May 2012**      Industrial Attachment, Survey of Kenya, Nairobi, Kenya.  
Undertook practicals in Geospatial Information Systems (GIS) by digitizing and cleaning the resultant data; Carried out cadastral survey computations and checking.
- Jan. 2011 – May 2011**      Industrial Attachment, Mapzone Company, Nairobi, Kenya.  
Acquired experience in the following areas: Cadastral Surveying, mainly in fieldwork, carrying out the computations and drawing using AutoCAD; Drawing of longitudinal profiles using software like TopoCAD and AutoCAD; Topographical Surveying, mainly carrying out the fieldwork, computations, and drawing topographical maps; Carrying out engineering surveys using total station, theodolite, and handheld GPS; and Used AutoCAD and LisCAD to draw longitudinal sections and cross-sections.
- Jan 2009 – May 2009**      Internal Attachment, Jomo Kenyatta University of Agriculture and Technology.  
Learned how to carry out traverses, leveling, and cadastral surveying as well as carrying out office work using AutoCAD, LisCAD, and TopoCAD. Instruments used included Electronic Distance Measuring Equipment, theodolite, and automatic & dumpy levels.

**Work Undertaken that Best Illustrates Capability to Handle the Tasks Assigned**

| From     | To       | Company/Project/Position/Relevant Technical & Management Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| Aug 2025 | Aug 2025 | <p><b>Name of Assignment:</b> Drone Training for Security Surveillance and Intelligence Operations Using DJI Matrice 350 RTK</p> <p><b>Client:</b> National Intelligence and Security Agency (NISA), Federal Republic of Somalia</p> <p><b>Main Project Features:</b> Capacity-building program designed to enhance NISA's operational capabilities in aerial surveillance, intelligence gathering, and situational awareness using the DJI Matrice 350 RTK drone system equipped with the Zenmuse H20T multi-sensor payload. The training covered flight planning, mission execution, payload operation (thermal, zoom, and wide sensors), and RTK positioning setup for precise data acquisition. Additional modules included data download, post-processing, geospatial analysis, and intelligence visualization for operational decision-making. The program adhered to DJI Enterprise flight safety protocols, KCAA and ICAO UAS operational standard and best practices for security-based drone deployment.</p> <p><b>Position Held:</b> Lead Trainer / Project Manager</p> <p><b>Activities Performed:</b> Designed and delivered theoretical and practical training modules; supervised hands-on flight exercises using the Matrice 350 RTK and H20T payload; guided participants in real-time mission setup, flight control, and emergency response procedures instructed on data processing workflows, including thermal imagery interpretation, mapping, and intelligence extraction; prepared training manuals and assessment materials and compiled a comprehensive training completion report and operational readiness recommendations for NISA's drone surveillance program.</p> |

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| July 2025 | Ongoing | <p><b>Name of Assignment:</b> Consultancy Services for Mutation Survey of Mwereni Group Ranch Phase II, Mwereni Ward</p> <p><b>Client:</b> Kwale County Government – Department of Environment &amp; Natural Resources</p> <p><b>Main Project Features:</b> Execution of a mutation survey and parcel boundary adjustment for Mwereni Group Ranch Phase II to support land subdivision, titling, and registration in compliance with the Survey Act (Cap 299) and Land Registration Act (2012) of Kenya. The assignment involved establishing control points, field boundary re-establishment, parcel mutation mapping, and data verification against existing cadastral records. The survey applied GNSS (RTK) and Total Station methods for accuracy and alignment with Survey of Kenya (SOK) geodetic standards. The project also included data cleaning, mutation plan preparation, validation with the Director of Surveys, and submission of final deliverables for approval and registration.</p> <p><b>Position Held:</b> Project Manager / Licensed Survey Coordinator</p> <p><b>Activities:</b> Oversee planning and coordination of fieldwork operations; established and verified geodetic control points; supervise boundary demarcation, GNSS observations, and mutation plan preparation; review data quality and ensured consistency with national cadastral standards; coordinated stakeholder and community consultations for boundary validation; prepare and submit final mutation survey plans and reports; and manage correspondence with the County Department of Environment &amp; Natural Resources and the Office of the Director of Surveys for approval.</p> |
| June 2025 | Ongoing | <p><b>Name of Assignment:</b> 3D Mapping and Deformation Monitoring of KeNHA Structures (Roads &amp; Bridges)</p> <p><b>Client:</b> Kenya National Highways Authority (KeNHA)</p> <p><b>Main Project Features:</b> 3D mapping and structural deformation monitoring exercise for selected KeNHA road and bridge infrastructure to support engineering assessment and maintenance planning. The project involved establishing a control network using GNSS survey techniques, performing 3D laser scanning using the Trimble X7 scanner, and integrating geospatial and engineering data for accurate deformation analysis. The assignment also included computation and submission of TS/ST forms for approval, creation of permanent survey monuments and markers, and generation of a detailed 3D geospatial model and technical report for structural condition evaluation.</p> <p><b>Position Held:</b> Project Manager / Lead Geospatial Engineer</p> <p><b>Activities Performed:</b> Supervised project planning, field execution, and data quality assurance; established survey control using GNSS receivers referenced to Survey of Kenya (SOK) known pillars; verified and transferred controls to site monuments; oversaw field scanning operations with the Trimble X7 laser scanner; coordinated point cloud processing, georeferencing, and deformation analysis; prepared and submitted TS/ST computations for approval; and compiled the final technical report detailing methodology, results, and recommendations for continuous monitoring and asset management.</p>                                                                                                  |
| Aug 2024  | Ongoing | <p><b>Name of Assignment:</b> Application of Drone Solutions for Sugarcane Mapping.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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|          |          | <p><b>Client:</b> Kenya Sugar Board (KSB)</p> <p><b>Main Project Features:</b> To establish a structured, transparent, and mutually beneficial framework and partnership for cooperation in promoting and building knowledge in geospatial technologies through research, education, training, outreach and consultancy in the areas of sugarcane mapping, remote sensing, yield prediction, digitization of field records, and staff capacity building.</p> <p><b>Position Held:</b> Project Manager and GIS Analyst</p> <p><b>Activities Performed:</b> To develop a yield predictive module for spatial imagery data processing and analysis. Build the staff human resource capacity on modern hands-on geospatial skills and competencies for the prevailing analysis tasks, and set up drone enabled data collection infrastructure at the board.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Dec 2023 | Aug 2024 | <p><b>Name of Assignment:</b> Consultancy to undertake the implementation of Shimoni Urban Development Plan.</p> <p><b>Client:</b> Kwale County Government – Department of Environment &amp; Natural Resources.</p> <p><b>Main Project Features:</b> Implementation of the Shimoni Urban Development Plan aimed at guiding sustainable urban growth, infrastructure development, and land use planning for Shimoni Town. The assignment involved acquisition of high-resolution drone imagery to support spatial planning and mapping, household enumeration for socio-economic profiling, and preparation of the Urban Development Plan in compliance with the Physical and Land Use Planning Act, 2019. The project emphasized participatory planning through community engagement, validation workshops, and integration of local stakeholder feedback into the plan. The final deliverables included the approved Urban Development Plan maps, reports, and implementation framework.</p> <p><b>Position Held:</b> Project Manager / Surveyor</p> <p><b>Activities Performed:</b> Coordinated drone aerial mapping and supervised data processing for base map generation; oversaw household enumeration exercise and socio-economic data analysis; led preparation of the first draft plan and facilitated stakeholder consultation meetings; incorporated community and institutional feedback into the second and final drafts; presented the final Urban Development Plan to the County Physical Planning Committee for adoption; and ensured quality assurance, documentation, and submission of approved deliverables.</p> |
|          |          | <p><b>Name of Assignment:</b> Ground Penetrating Radar (GPR) Mapping of the Old Mombasa/Uhuru Highway</p> <p><b>Client:</b> Kenya National Highways Authority (KeNHA)</p> <p><b>Main Project Features:</b> Execution of a Ground Penetrating Radar (GPR) mapping and subsurface utility detection survey along the Mombasa/Uhuru Highway corridor to support rehabilitation, expansion, and structural improvement works. The project aimed at identifying, mapping, and profiling underground utilities and subsurface anomalies, including drainage lines, electrical conduits, and voids, to inform design and engineering decisions. The assignment combined GPR data acquisition, GNSS-based georeferencing,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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|           |         | <p>and GIS integration for the creation of a unified digital subsurface database. The processed GPR data were used to generate detailed cross-sectional profiles, depth interpretations, and subsurface infrastructure maps in alignment with KeNHA's survey and engineering data standards.</p> <p><b>Position Held:</b> Project Manager / Lead Geospatial Engineer</p> <p><b>Activities Performed:</b> Directed project planning and coordination with KeNHA engineers and contractors; supervised GPR field data acquisition and survey control establishment using survey grade GNSS receivers; ensured accurate positioning and referencing of GPR transects; oversaw data processing, filtering, and interpretation for subsurface feature detection; Involved in integration of GPR results into CAD environments for visualization and engineering design use; reviewed and validated deliverables; and prepared the final technical report and GPR interpretation maps submitted to KeNHA for project implementation and design integration.</p>                             |
| Jan 2024  | Ongoing | <p><b>Name of Assignment:</b> Development of ProSync (CRM/ERP Application)</p> <p><b>Client:</b> Ecospace Services Limited</p> <p><b>Main Project Features:</b> Design and development of a comprehensive CRM/ERP system integrating client management, invoicing, document management, workflow automation, and financial reporting modules. The project also includes integration with Mayan EDMS for document handling, KCB for payment automation, and PostgreSQL backend database for unified data management.</p> <p><b>Position Held:</b> Project Manager and UI/UX Designer</p> <p><b>Activities Performed:</b> Supervised the overall system design, API architecture, and Database structure; guided UI/UX development for both web and mobile interfaces; reviewed integration with external services (Mayan EDMS and KCB Buni); conduct system testing and user acceptance trials, developed user manuals, and oversaw deployment and maintenance strategies for sustainable operation.</p>                                                                               |
| June 2023 | Ongoing | <p><b>Name of Assignment:</b> Provision of Consultancy Services for the Planning of Colonial Villages (Ichamara &amp; Itundu).</p> <p><b>Client:</b> Nyeri County Government: Department of Lands, Physical Planning &amp; Urban Development.</p> <p><b>Main Project Features:</b> Preparation of a detailed local physical development plans for the colonial villages of Ichamara and Itundu in Nyeri County, aimed at guiding land regularization and infrastructure development. The assignment involves the use of Drone technology to acquire high-resolution imagery for accurate base mapping and planning, coupled with household enumeration to capture socio-economic and land occupation data. The process included stakeholder engagement, community validation, and preparation of successive plan drafts in line with Kenya's Physical and Land Use Planning Act, 2019. The final deliverables include approved development plans, spatial datasets, and policy recommendations to support implementation.</p> <p><b>Position Held:</b> Survey and Project Manager</p> |

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|           |             | <p><b>Activities Performed:</b> Directed and supervised drone data acquisition and orthomosaic generation, oversaw household enumeration and data validation; led spatial analysis, plan preparation, and community participation sessions; presented the first draft plan to stakeholders, incorporated community feedback into the second draft, and coordinated the final approval process. Successfully completed and submitted the Ichamara Village Plan, with Itundu Village Plan currently under finalization.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Dec, 2022 | March, 2023 | <p><b>Name of Assignment:</b> Proposed Subdivision into two Agricultural portion of L/R No. 30443-9 South West of Lake Naivasha</p> <p><b>Client:</b> OSERIAN DEVELOPMENT CO LTD</p> <p><b>Main Project Features:</b> Geodetic Control transfer to the site, Physical planning, submission of survey scheme to the county offices for approvals, carry out Land Sub-division at La-Pieve Farm, Naivasha Sub-County</p> <p><b>Position Held:</b> Surveyor</p> <p><b>Activities Performed:</b> Data collection, Geodetic control transfer, subdivision survey, data processing, computation and Report writing.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| June 2022 | Ongoing     | <p><b>Name of Assignment:</b> Development of a Land Information Management System (LIMS) for Turkana County Government</p> <p><b>Client:</b> Turkana County Government – Department of Lands, Physical Planning, and Urban Development</p> <p><b>Main Project Features:</b> Design and development of an integrated Land Information Management System (LIMS). The system consolidates cadastral data, parcel ownership records, and land use information into a centralized geospatial database. It incorporates modules for land registration, mutation management, planning applications, and automated title processing workflows.</p> <p><b>Position Held:</b> Project Manager / Geospatial Applications Engineer.</p> <p><b>Activities Performed:</b> Led the technical design and system architecture development, coordinated the creation of the geodatabase and sample data migration, guided the development of the UX/UI, supervised backend API development, spatial data integration, and web mapping functionalities ensuring adherence to land governance standards and data security protocols, conducted stakeholder training and application progress testing.</p> |
| Oct, 2022 | Dec, 2022   | <p><b>Name of Assignment:</b> Supply and Delivery of Drone Images for selected informal settlements in Kenya.</p> <p><b>Client:</b> State Department for Housing and Urban Development – Second Kenya Informal Settlements Improvement Project (KISIP 2)</p> <p><b>Main Project Features:</b> Control Transfer and GCP Densification and Drone Mapping.</p> <p><b>Position Held:</b> Surveyor and Drone Pilot</p> <p><b>Activities Performed:</b> Control Transfer and GCP Densification, GNSS Data processing, Drone Mapping and Drone Data processing.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Jan 2022  | March 2023  | <p><b>Name of Assignment:</b> Proposed Subdivision into five portions for Agricultural, Residential, Commercial, and Tourism purposes of L/R No. 7426-11 South West of Lake Naivasha</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



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|           |           | <p><b>Client:</b> La-Pieve Limited</p> <p><b>Main Project Features:</b> Geodetic Control transfer to the site, Physical planning, submission of survey scheme to the county offices for approvals, carry out Land Sub-division at La-Pieve Farm, Naivasha Sub-County</p> <p><b>Position Held:</b> Surveyor</p> <p><b>Activities Performed:</b> Data collection, Geodetic control transfer, subdivision survey, data processing, computation and Report writing.</p>                                                                                                                                                                                                                                                             |
| Dec, 2021 | May, 2022 | <p><b>Name of Assignment:</b> Development of Land Cover Mobile Application</p> <p><b>Client:</b> Kenya Space Agency</p> <p><b>Main Project Features:</b> Mobile App Overall UI/UX design and development, Web App UI/UX design and development, API development, Integration &amp; testing, Database design, development, integration and testing, System review and Implementation</p> <p><b>Position Held:</b> Project Manager</p> <p><b>Activities Performed:</b> Mobile App Overall UI/UX design, Web App UI/UX design, Mobile and Web App testing, and user manual development.</p>                                                                                                                                        |
| Nov, 2020 | Jan, 2021 | <p><b>Name of Assignment:</b> Subdivision of L/R No. 7426-5 La-Pieve</p> <p><b>Client:</b> OSERIAN DEVELOPMENT CO LTD</p> <p><b>Main Project Features:</b> Geodetic Control transfer to the site, Physical planning, submission of survey scheme to the county offices for approvals, carry out Land Sub-division at La-Pieve Farm, Naivasha Sub-County</p> <p><b>Position Held:</b> Surveyor</p> <p><b>Activities Performed:</b> Data collection, Geodetic control transfer, subdivision survey, data processing, computation and Report writing.</p>                                                                                                                                                                          |
| Oct, 2020 | Feb, 2021 | <p><b>Name of Assignment:</b> Cadastral Survey and Drone Mapping at Dhusamareeb.</p> <p><b>Client:</b> MATGURI DEVELOPERS LIMITED</p> <p><b>Main Project Features:</b> Geodetic control establishment from existing International GNSS Service stations, control densification, land planning and subdivision, GCP establishment, Flight planning, UAS mapping, data processing, and analysis. Preparation of contours and orthomosaic maps using ArcGIS for report writing, presentation, and discussion.</p> <p><b>Position Held:</b> Drone Pilot, Data Analyst and Surveyor.</p> <p><b>Activities Performed:</b> Data collection, staking out by RTK, Drafting using AutoCAD, Report writing, printing and presentation.</p> |
| Nov, 2020 | Nov, 2020 | <p><b>Name of Assignment:</b> Drone Mapping of Gogo Farm.</p> <p><b>Client:</b> ORTHOMAP SOLUTION LTD</p> <p><b>Main Project Features:</b> Geodetic control transfer from existing National Control points to enable us to establish the GCP, Flight planning, UAS mapping, data processing, and analysis. Preparation of contours and orthomosaic maps, report writing, presentation, and discussion</p> <p><b>Position Held:</b> Drone Pilot and Data Analyst</p>                                                                                                                                                                                                                                                             |

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|            |            | <b>Activities Performed:</b> Data collection, Geodetic control transfer, drone observers, data processing, analysis and Report writing.                                                                                                                                                                                                                                                                                                                                                |
| May 2020   | May 2020   | <b>Name of Assignment:</b> Demarcation of the Masinga Dam Boundary beacons<br><b>Client:</b> Tana and Athi River Development Authority (TARDA)<br><b>Main Project Features:</b> Control transfer/Densification, Beaconing, Computations and Map preparations;<br><b>Position Held:</b> Surveyor<br><b>Activities Performed:</b> Control Transfer, Data computation, beaconing, map preparation and report writing.                                                                     |
| April 2020 | April 2020 | <b>Name of Assignment:</b> Technical Support Services: Configuration of Trimble R10 GNSS receiver as NTRIP Server;<br><b>Client:</b> Seline Consultant, Kilifi.<br><b>Main Assignment Features:</b> Receiver updates, Configuration and Testing;<br><b>Position Held:</b> Geospatial Engineer<br><b>Activities Performed:</b> Receiver updates, created and configured an NTRIP caster and server, equipment data collection/testing using NTRIP as a mode of correction transmission. |
| Feb 2020   | Feb 2020   | <b>Name of Assignment:</b> Training on Fundamentals of Total Stations<br><b>Client:</b> Fujita Corporation, Mombasa<br><b>Main Assignment Features:</b> Introduction, Data Capture, Coordinate Systems, Data Download, Staking and station setup;<br><b>Position Held:</b> Trainer<br><b>Activities Performed:</b> Survey data collection using Total station, Data cleaning and Importing, Staking, Station setup and leveling using Digital level;                                   |
| July 2019  | Aug 2019   | <b>Name of assignment:</b> Consultancy to Conduct Topographical survey and setting out exercise at Aga Khan Hospital Nairobi.<br><b>Client:</b> Property Development Management (PDM)<br><b>Main Project Features:</b> Transfer and densification of Ground Control Points (GCPs), Detail Picking and Setting out;<br><b>Position held:</b> Surveyor<br><b>Activities Performed:</b> Field Data Collection and Computation, Data Processing, Drafting, setting out and Report Writing. |
| May 2019   | May 2019   | <b>Name of Assignment:</b> Installation of the Continuous Operating Reference System (CORS) Stations<br><b>Client:</b> Oakar Services<br><b>Main Assignment Features:</b> Antenna Mounting, Transporting Equipment to different locations, Setup and Configuration, Static Survey observations;<br><b>Position Held:</b> Geospatial Engineer<br><b>Activities Performed:</b> Setup and Configuration, Static Survey observations.                                                      |
| Feb 2019   | Feb 2019   | <b>Name of Assignment:</b> Training on Introduction to Geographic Information Systems (GIS)                                                                                                                                                                                                                                                                                                                                                                                            |

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|             |             | <p><b>Client:</b> Kwale County Survey and GIS department</p> <p><b>Main Assignment Features:</b> Introduction, Data Capture, Coordinate Systems, Data Visualization, Map Making, Database creation and integration to survey data;</p> <p><b>Position Held:</b> Trainer</p> <p><b>Activities Performed:</b> Survey data collection using GPS, Data cleaning and Importing, Database Creation, data digitization, data analysis and visualization, map production.</p>                                                                                                     |
| Aug 2018    | Dec. 2018   | <p><b>Name of Assignment:</b> Geospatial Database development and data capture</p> <p><b>Client:</b> Locomizer</p> <p><b>Main Project Features:</b> Development of Geospatial Database, Data capture, workflow model development, map production, web GIS development, data analysis and visualization.</p> <p><b>Position Held:</b> GIS &amp; Quality Assurance Consultant</p> <p><b>Activities Performed:</b> database development, data digitization, data analysis and visualization, map production, data validation and quality assessment, reports generation.</p> |
| Oct. 2018   | Oct. 2018   | <p><b>Name of assignment:</b> Consultancy to Map and Conduct Cadastral Survey services.</p> <p><b>Client:</b> Tana and Athi Water Development Authority (TARDA).</p> <p><b>Main Project Features:</b> Transfer and densification of Ground Control Points (GCPs), Mapping the Masinga Dam and Land subdivision.</p> <p><b>Position held:</b> Surveyor</p> <p><b>Activities Performed:</b> Field Data Collection and Computation, Data Processing, Drafting, Quality Control and Report Writing.</p>                                                                       |
| April. 2018 | April. 2018 | <p><b>Name of assignment:</b> Consultancy to Conduct Land sub-division Cadastral Survey work in Olkaria, Naivasha.</p> <p><b>Client:</b> Oserian Development Co. Ltd</p> <p><b>Main Project Features:</b> Transfer and densification of Ground Control Points (GCPs), Land subdivision.</p> <p><b>Position held:</b> Surveyor</p> <p><b>Activities Performed:</b> Field Data Collection and Computation, Data Processing, Drafting, Quality Control and Report Writing.</p>                                                                                               |
| Oct. 2017   | Dec. 2017   | <p><b>Name of assignment:</b> Consultancy to Conduct Engineering survey work for construction of power plant in Olkaria, Naivasha.</p> <p><b>Client:</b> Civicon Engineering</p> <p><b>Main Project Features:</b> Establishment of Ground Control Points (GCPs), Plant Setting Out and Levelling.</p> <p><b>Position held:</b> Surveyor</p> <p><b>Activities Performed:</b> Field Data Collection and Computation, Data Processing, Quality Control and Report Writing.</p>                                                                                               |
| Aug. 2017   | Nov. 2017   | <p><b>Name of assignment:</b> Consultancy to Conduct Road Construction Engineering survey work in Kisii</p> <p><b>Client:</b> Civicon Engineering</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |

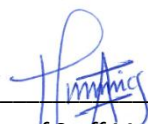
|           |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |           | <p><b>Main Project Features:</b> Establishment of Ground Control Points (GCPs), Road Setting Out and Levelling.</p> <p><b>Position held:</b> Surveyor</p> <p><b>Activities Performed:</b> Field Data Collection and Computation, Data Processing, Quality Control and Report Writing.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Jun. 2017 | Aug. 2017 | <p><b>Name of assignment:</b> Consultancy to Conduct Setting out Engineering Survey work for the Kigoro water treatment plant.</p> <p><b>Client:</b> Sogea-Satom</p> <p><b>Main Project Features:</b> Establishment of Ground Control Points (GCPs), Setting Out and Levelling.</p> <p><b>Position held:</b> Surveyor</p> <p><b>Activities Performed:</b> Field Data Collection and Computation, Data Processing, Quality Control and Report Writing.</p>                                                                                                                                                                                                                                                                                                                                    |
| Feb. 2017 | Feb. 2017 | <p><b>Name of assignment:</b> Establishment of Ground Control Point (GCP) Project that involved Control survey to establish Ground Control Points (GCPs) within a specified area of interest in Nyanza, Rwanda to be used to geo-reference satellite imagery for generation of Digital Terrain Model.</p> <p><b>Client:</b> Mawarid Rwanda Ltd</p> <p><b>Main Project Features:</b> Establishment of Ground Control Points (GCPs)</p> <p><b>Position held:</b> Surveyor</p> <p><b>Activities Performed:</b> Field Data Collection and Computation, Data Quality Control and Report Writing.</p>                                                                                                                                                                                              |
| Feb. 2015 | Feb. 2017 | <p><b>Name of Assignment:</b> Consultancy Services for the Design, Development and Implementation of a Geographical Information System-Second Phase for UETCL</p> <p><b>Client:</b> Uganda Electricity Transmission Company Ltd (UETCL)</p> <p><b>Funded By:</b> World Bank</p> <p><b>Main Project Features:</b> User Needs Assessment, business process analysis, workflow model development, data model development, web GIS development, data collection, Training and Capacity Development as well as setting up of the GIS lab.</p> <p><b>Position Held:</b> GIS Analyst/Instructor</p> <p><b>Activities Performed:</b> User Needs Assessment, Software Installations and Configuration, Project Report Writing, Data Editing, Conducting Training and Technical Support to Client.</p> |
| Jan. 2017 | Feb. 2017 | <p><b>Name of Assignment:</b> Consultancy to Conduct Setting out Engineering survey work</p> <p><b>Client:</b> Elemtech Engineering</p> <p><b>Main Project Features:</b> Control transfer and densification, setting out high voltage power transmission lines.</p> <p><b>Position Held:</b> Surveyor</p> <p><b>Activities Performed:</b> Software Installations and Configuration, Project Report Writing, Conducting Training and Technical Support to Client.</p>                                                                                                                                                                                                                                                                                                                         |

|             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nov, 2016   | Nov, 2016   | <p><b>Name of assignment:</b> Topographic Survey of a plot in development adjacent to the Nairobi West Hospital. Project entailed picking all the features on the plot and adjacent to the plot in order to generate contour lines and all the properties of the area.</p> <p><b>Client:</b> Nairobi West Hospital</p> <p><b>Main Project Features:</b> Topographic Survey</p> <p><b>Position held:</b> Surveyor</p> <p><b>Activities Performed:</b> Field Data Collection and Computation, Data Quality Control, Drawing and Report Writing.</p>                                                                                    |
| March, 2015 | May, 2016   | <p><b>Name of Assignment:</b> Supply of Geographic Information System to the Kenya National Highways Authority (KeNHA)</p> <p><b>Client:</b> Kenya National Highways Authority (KeNHA)</p> <p><b>Funded By:</b> World Bank</p> <p><b>Main Project Features:</b> Supply and Installation of a GIS hardware, software, Importation of Kenya Roads Board dataset to KeNHA GIS and training and capacity building</p> <p><b>Position Held:</b> GIS Analyst</p> <p><b>Activities Performed:</b> Overseeing Data Editing, Preparation of Customized Training Materials, Creation of a GIS Multiuser Geodatabase and Technical Support.</p> |
| March, 2015 | Feb., 2016  | <p><b>Name of Assignment:</b> Provision of Geographical Information System Data Collection Services.</p> <p><b>Client:</b> Kenya Power &amp; Lighting Company</p> <p><b>Main Project Features:</b> GPS Data Collection of Low Voltage Networks and Customer Meters, Data Verification and Editing on Facility Database (FDB).</p> <p><b>Position Held:</b> Project Supervisor</p> <p><b>Activities Performed:</b> Supervising 50 data collectors, Field Data Collection, Data Verification and Quality Control, Report Writing and Training on GPS data collection.</p>                                                              |
| July, 2014  | March, 2015 | <p><b>Name of Assignment:</b> Development of a GIS and Knowledge Management Centre for Arid and Semi-Arid Lands (ASAL)</p> <p><b>Client:</b> Ministry of Devolution and Planning, State Department of Devolution,</p> <p><b>Main Project Features:</b> User Needs Assessment, GIS development, Training and Capacity Development as well as setting up of the GIS center.</p> <p><b>Position Held:</b> Team Leader and GIS Instructor</p> <p><b>Activities Performed:</b> Field Data Collection and Compilation, Map Production, Report Writing and Training and Capacity Development on ArcGIS and Web Mapping.</p>                 |
| Sep, 2013   | Feb., 2014  | <p><b>Name of Assignment:</b> GIS installation &amp; Setup at the NIB</p> <p><b>Client:</b> National Irrigation Board (NIB)</p> <p><b>Main Project Features:</b> Supply and Installation of a GIS for the Management of the Irrigation Schemes including Hardware, Software, Imagery and Training and Capacity Building</p> <p><b>Position Held:</b> Team Leader</p>                                                                                                                                                                                                                                                                 |

|           |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |             | <b>Activities Performed:</b> Data Development and Compilation, Cartographic Map Production, Technical Support and Training on GIS, Remote Sensing and GPS Data Collection.                                                                                                                                                                                                                                                                                                                    |
| May, 2014 | Jun., 2014  | <b>Name of assignment:</b> Setting Out the transmission Power line of 132KV from Konza/Machakos to Namanga<br><b>Client:</b> Shymar Power India Limited Main<br><b>Project Features:</b> To transfer controls and fix the center line. Position held: Surveyor<br><b>Activities Performed:</b> Transferred control via static survey, carried out topographical survey of the proposed substation and fixed beacons along the proposed HV line.                                               |
| Nov, 2013 | Aug., 2014  | <b>Name of Assignment:</b> GIS Consultancy Service for the Mapping of Guinness Nigeria Outlets in Lagos<br><b>Client:</b> Tylor Nelson Sofres (TNS) and Research and Marketing Services (RMS)<br><b>Main Project Features:</b> Data Cleanup and Verification, GIS development as well as GIS Software Delivery and Installation.<br><b>Position Held:</b> GIS Analyst<br><b>Activities Performed:</b> Data Cleanup, Verification and Compilation and Map Production                           |
| Nov, 2013 | April, 2014 | <b>Name of Assignment:</b> GIS Consultancy Service for the Mapping of Newspaper Vendors within Nairobi County.<br><b>Client:</b> Nation Media Group<br><b>Main Project Features:</b> Data Collection, Data Cleanup and Verification, GPS Training, GIS Database Development, QGIS Training and Map Production.<br><b>Position Held:</b> Team Leader and Cartographer<br><b>Activities Performed:</b> Data Cleanup, Verification and Compilation, Training, Map Production and Report Writing. |
| Jun, 2014 | Jun, 2014   | <b>Name of assignment/Job or project:</b> Training the Technical University of Kenya (TUK) students on GNSS equipment<br><b>Client:</b> Technical University of Kenya (TUK)<br><b>Position held:</b> Trainer<br><b>Activities performed:</b> Taught the students how to setup the equipment, do static and RTK survey. This also included how to compute and analyse the data collected                                                                                                       |

### 13. Certification

I, the undersigned, certify that to the best of my knowledge and belief that this CV correctly describes myself, my qualifications, and my experience. I understand that any wilful misstatement described herein may lead to my disqualification or dismissal, if engaged.

  
\_\_\_\_\_  
[Signature of Staff Member]

Date: 13/10/2025  
Day/Month/Year